

# Barath R

## Embedded Engineer

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### PROFESSIONAL SUMMARY

Third-year Electronics and Communication Engineering student and Embedded Engineer with hands-on proficiency across hardware design, bare-metal firmware, and software engineering. Experienced in building from silicon to software -- from register-level RISC-V assembly and custom PCB/sensor bring-up to full-stack IoT telemetry pipelines, cloud integration, and algorithmic problem solving. Seeking an Embedded Engineer role to architect high-performance, connected hardware and software systems.

### EDUCATION

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| <b>B.E. Electronics and Communication Engineering</b><br>Rajalakshmi Institute of Technology, Chennai<br><i>Currently in 3rd year</i>                     | 2024 -- 2028 (Expected)<br><b>CGPA: 8.65/10</b> |
| <b>Higher Secondary Certificate (HSC) -- Class XII</b><br>Montfort Matriculation Higher Secondary School, Chennai<br><i>Tamil Nadu State Board</i>        | 2023 -- 2024<br><b>Percentage: 95%</b>          |
| <b>Secondary School Leaving Certificate (SSLC) -- Class X</b><br>Montfort Matriculation Higher Secondary School, Chennai<br><i>Tamil Nadu State Board</i> | 2021 -- 2022<br><b>Percentage: 94%</b>          |

### TECHNICAL SKILLS

**Hardware & Embedded Systems:** ESP32, ESP32-C6, ESP32-CAM, Raspberry Pi 5, Arduino UNO R3, Altera DE2 FPGA, Bare-Metal MMIO, CST Studio, Keysight ADS

**PCB Design & EDA Tools:** PCB/PCD Design, KiCad, Proteus, EasyEDA, Schematic Capture, Multi-Layer PCB Layout, Routing, DRC/ERC, Gerber & BOM Generation, Component Footprints, SMD/THT Soldering, Hardware Bring-up

**Hardware Protocols & Interfacing:** I2C, SPI, UART, MQTT, LoRaWAN, GPIO, PWM, ADC, MMIO, WebSockets, HTTP/REST

**Programming Languages:** C, C++, Python, JavaScript (ES6+), TypeScript, Java, SQL, RISC-V Assembly

**Software & Web Development:** React.js, Next.js, Node.js, Express.js, FastAPI, Flask, HTML5, CSS3, Tailwind CSS, REST APIs

**Cloud & Databases:** Firebase, HiveMQ, ThingSpeak, AWS IoT Core, Google Cloud Platform, MongoDB, PostgreSQL, MySQL, SQLite

**Engineering Tools & DevOps:** Git, GitHub Actions, Docker, Linux/Bash, ESP-IDF (v6), VS Code, PlatformIO, MATLAB, Scilab, ADB

**Core CS & Problem Solving:** Data Structures & Algorithms (LeetCode streak), OOP, Concurrency, RTOS Architecture

### FEATURED PROJECTS

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| <b>Bare-Metal RISC-V Trap Handler &amp; Context Switcher</b><br><i>Tech Stack: RISC-V Assembly, C, Direct MMIO, Hardware Timers, Interrupt Controller</i> <ul style="list-style-type: none"><li>Designed and implemented a bare-metal trap handler and timer-based preemptive context switcher in assembly with no RTOS or vendor libraries</li><li>Configured memory-mapped I/O (MMIO) registers for hardware interrupt handling, stack frame preservation, and register-level task scheduling</li></ul>                   | ESP32-C6                 |
| <b>Digital Twin for Industrial Monitoring</b><br><i>Tech Stack: ESP32 / Edge Node, Multi-Sensor Fusion, 5G Telemetry, Python, Real-Time Web Dashboard</i> <ul style="list-style-type: none"><li>Architected an industrial digital twin connecting multi-sensor edge nodes with 5G connectivity for real-time equipment telemetry and predictive maintenance</li><li>Developed cloud data ingestion pipelines and responsive dashboard visualization, bridging sensor-level data acquisition to software analytics</li></ul> | 5G-Integrated IoT System |
| <b>Dual-Axis Solar Tracker</b><br><i>Tech Stack: Custom PCB Prototyping, Microcontroller, Servo Motors, LDR Sensor Array, Closed-Loop Control, C/C++</i>                                                                                                                                                                                                                                                                                                                                                                    | Embedded Control System  |

- Constructed a dual-axis solar tracker optimizing photovoltaic panel orientation using an analog LDR sensor bridge and closed-loop feedback
- Designed custom sensor interfacing and motor driver circuitry with voltage regulation; programmed precision tracking algorithms in C/C++

### **Conformal Helmet-Integrated Antenna Array**

*RF & Antenna Hardware*

*Tech Stack: Microstrip Patch Array, Dual-Band UHF/L-Band, Flexible PCB Substrate, CST Microwave Studio, ADS, RF Shielding*

- Engineered a flexible microstrip patch antenna array conformal to curved helmet geometry, replacing bulky whip antennas for tactical gear
- Modeled dual-band UHF/L-band radiation patterns in CST/ADS and integrated an RF shielding layer with ruggedized coax interfacing

### **TECHNICAL EXPERIENCE**

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- Conceptualized, engineered, and deployed embedded and IoT solutions end-to-end, from custom PCB design (schematic to fabrication) and board bring-up to bare-metal firmware and full-stack dashboards
- Executive Technical Organizer at IETE Students' Forum (ISF), coordinating technical hackathons and workshops across software and hardware tracks

### **LANGUAGES**

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English (Fluent) • Tamil (Native)